

30.3 A 28nm 32Kb Embedded 2T2MTJ STT-MRAM Macro with 1.3ns Read-Access Time for Fast and Reliable Read Applications

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Many IoT and wearable devices require an on-chip small-to-mid-capacity nonvolatile memory (NVM) with a fast read-access time (T_{AC}) and reliable read operations: for applications including data-logging, configurable look-up tables (LUT), eFuse, and physically unclonable functions (PUF). STT-MRAM [1-4] is a good candidate for these applications due to its fast write speed, low-voltage write, and high endurance. However, STT-MRAM suffers from a small read-signal margin (RSM) due to a small tunnel magnetoresistance ratio (TMR: $(R_{AP}-R_P)/R_P$) between the cell resistance of parallel (P, R_P) and anti-parallel (AP, R_{AP}) states [1-6]. Moreover, the read-disturb behavior of STT-MRAM cells is sensitive to the BL read voltage (V_{BL_RD}) and the stress/read time. Compact 1T1MTJ arrays are suitable for high-density applications [5-6]; however, they use a power-hungry current-mode read scheme with a slow read speed due to the small RSM. Researchers have proposed 2T2MTJ (Fig. 30.3.1) arrays [1-4] with differential bitlines (BL and BLB) and a voltage-mode read scheme, with an enlarged RSM (V_{RSM}), for fast, low-power read operations. V_{RSM} refers to the voltage difference between BL (V_{BL}) and BLB (V_{BLB}). 2T2MTJ STT-MRAM read operations still face the following challenges: (1) V_{BL} and V_{BLB} both drop from V_{BL_RD} to 0V quite quickly due to the large cell read current (I_P and I_{AP}) or low R-value in both R_P and R_{AP} , resulting in small sensing window (T_{SMW}), which is the period when $V_{RSM} > \text{offset}$; (2) the maximum V_{RSM} (V_{RSM_MAX}) occurs at different times (t_{RSM_MAX}) for different cells due to TMR (R_{AP}/R_P) variation; and (3) a degraded V_{RSM} due to the use of a low V_{BL_RD} to avoid read disturbs for high data-reliability applications. (1) and (2) lead to a decrease in V_{RSM} after reaching its peak (V_{RSM_MAX}), despite an increase in BL development time (t_{BL}). When using a conventional voltage-mode sense amplifier (CNV-VSA) with a common activated (SAEN=1) timing (t_{SAEN}) under the effects of (1)-(3), the signal to be amplified ($\Delta V_{IN} < V_{RSM_MAX}$) is subject to degradation at the VSA's differential inputs, resulting in a sensing failure at a low V_{BL_RD} .

A continuous-recording-and-enhancement (CRE) VSA, shown in Fig. 30.3.2, is proposed, which can tolerate a small TMR-Ratio to achieve a fast T_{AC} at a low V_{BL_RD} , and achieves a reduced read power. The fabricated 28nm 32kb 2T2MTJ STT-MRAM macro achieves the fastest read-access time, 1.3ns, among recently reported STT-MRAMs.

The CRE-VSA employs a CRE unit with a conventional latch-based VSA (LVSA) for margin enhancement and offset suppression, due to V_{TH} variation in the input stage. This is accomplished by continuously recording the V_{RSM} over time and increasing ΔV_{IN} as t_{BL} increases, despite a corresponding decrease in V_{RSM} as t_{BL} increases.

The detailed operation of the proposed CRE-VSA is shown in Fig. 30.3.3 and 30.3.4: each CRE unit includes a pair of capacitors (C1 & C2), cross-coupled transistors (M3 & M4), precharge transistors (M1 & M2) and reset transistors (M5 & M6). The CRE-VSA operates in three phases (P1-P3). In the standby state – starting at the end of a read operation to the beginning of the next – M5 and M6 are on to reset V_{X1} and V_{X2} to 0V, while PRE=1 (M1 & M2 are off) and V_{IN1} and V_{IN2} is 0V. In P1 (V_{TH} sampling and BL precharge), M5 and M6 are off, and M1 and M2 are on to precharge V_{IN1} and V_{IN2} to V_{DD} , such that $V_{X1}=V_{DD}-V_{TH3}$ and $V_{X2}=V_{DD}-V_{TH4}$. This results in M3 and M4 biased at the edge of on-off boundary. At the same time, the V_{BL} and V_{BLB} are precharged to V_{BL_RD} – similar to a conventional read scheme – so that P1 has no timing overhead. In P2 (BL development and continuous recording and margin enhancement), M1 and M2 are off, such that IN1 and IN2 are floating. When the WL is on, BL and BLB are discharged by the cell read current (I_P & I_{AP}). The voltage swing (V_S) on BL and BLB ($V_{S_BL}=V_{BL_RD}-V_{BL}$ & $V_{S_BLB}=V_{BL_RD}-V_{BLB}$) is coupled to nodes X1 and X2 via C1 and C2, which results in $V_{X1}=V_{DD}-V_{TH3}-V_{S_BL}$ and $V_{X2}=V_{DD}-V_{TH4}-V_{S_BLB}$. M3 and M4 are then turned on, such that the over-drive voltage of M3 and M4 becomes $V_{OD3}=V_{IN2}-V_{DD}+V_{S_BL}$ and $V_{OD4}=V_{IN1}-V_{DD}+V_{S_BLB}$, which is independent of V_{TH} variation. V_{IN1} and V_{IN2} then begin to drop in accordance with the discharge drain current (I_{D3} and I_{D4}) or the charge sharing current between X1-IN1 and X2-IN2, as controlled by M3 and M4 (V_{OD3} & V_{OD4}). With an increase in t_{BL} , the V_{S_BL} and V_{S_BLB} of each time-step contributes to a differential voltage drop (ΔV_S) at X1 and

X2, which continues to affect V_{IN1} and V_{IN2} , V_{OD3} and V_{OD4} , and I_{D3} and I_{D4} . The cross-coupled feedback between M3 and M4 (V_{OD3} & V_{OD4}) causes an increase in the drain current ($V_{OD} > V_{RSM} > 0$) of the CRE path (C1-M3-IN1 & C2-M4-IN2) with a larger V_S . At the same time, the other path (with a smaller V_S) sees a decrease in drain current, which approaches zero when M3 and M4 move into the sub-threshold region ($V_{OD} < 0$). As a result, the voltage difference between V_{IN1} and V_{IN2} (ΔV_{IN}) is k-times V_{RSM} . Moreover, an increase in t_{BL} leads to a constant increase in ΔV_{IN} , despite a drop in V_{RSM} after it reaches a peak. This trend differs from that of conventional VSA.

When the BL starts (t_1) to develop for read-1 case ($V_{S_BL} > V_{S_BLB}$ or $V_{BL} < V_{BLB}$) the fact that $V_{X1} < V_{X2}$ results in $V_{IN1} < V_{IN2}$ and $V_{OD3} > V_{OD4}$. This causes $I_{D3} > I_{D4}$, which moderates the rate at which V_{IN2} drops. Later ($@t_2$), the voltage drop at IN2 ($\Delta V_{IN2}=V_{DD}-V_{IN2}$) is significantly smaller than V_{S_BL} , while the voltage drop at IN1 ($\Delta V_{IN1}=V_{DD}-V_{IN1}$) is larger than V_{S_BLB} . As a result, M3 is in the saturation region and M4 is in the sub-threshold region. C1 continuously records/couples V_{S_BL} to X1 to lower V_{IN1} , whereas V_{IN2} undergoes no voltage drop – keeping M4 in cutoff – despite a continually increasing V_{S_BLB} . V_{BLB} approaches 0V and V_{X2} continues dropping. Furthermore, ΔV_{IN} keeps increasing from t_1 to reach a maximum value (ΔV_{IN_MAX}) when V_{BL} reaches 0V. Then, ΔV_{IN} remains at ΔV_{IN_MAX} even as V_{RSM} decreases: V_{BLB} keeps dropping toward 0V.

Figure 30.3.5 presents the performance of the CRE-VSA. Unlike a conventional VSA, the ΔV_{IN} of the CRE-VSA increases with an increase in t_{BL} . The maximum (ΔV_{IN_MAX}) of the CRE-VSA is 5.85 $\times$ larger than that of a conventional VSA. The small offset and enhanced margin of the CRE-VSA provides tolerance for a minimum TMR-ratio (TMR_{MIN}) that is 5.6 to 8 $\times$ smaller than that of a conventional VSA across various resistance values. With the same sensing yield, the CRE scheme can tolerate a V_{BL_RD} that is 2 $\times$ lower than that of a conventional VSA, which reduces the read-disturb rate by more than 10 $\times$. At different BL lengths, this scheme allows for a 1.5-1.86 $\times$ shorter T_{AC} than a conventional VSA, when TMR_{MIN}=100% and V_{BL_RD} =0.3V. A lower V_{BL_RD} reduced the read energy of the CRE-VSA by 34%, compared to a conventional VSA with the same TMR_{MIN} sensing yield.

A 28nm 32kb 2T2MTJ macro is fabricated with test-modes using a CRE-VSA and a conventional VSA, using similarly skewed transistor sizes in the latch circuit. Figure 30.3.6 presents the measurement results. For a 32kb macro with limited TMR-ratio variation across cells, CRE achieves a 1.3ns macro-level read-access time (T_{AC_MACRO}) at a V_{BL_RD} =0.3V, which is 1.43 $\times$ faster than a conventional VSA where T_{AC_MACRO} is 1.86ns. A larger macro size and a higher TMR variation is expected to increase the T_{AC} reduction ratio. Moreover, the CRE scheme works at V_{BL_RD} =140mV, which is 2 $\times$ lower than a conventional VSA (V_{BL_RD} =280mV) at room temperature. At higher temperatures (e.g. 75°C), the improvement in minimum V_{BL_RD} increases to 190mV. Figure 30.3.7 presents the die photo and a chip summary.

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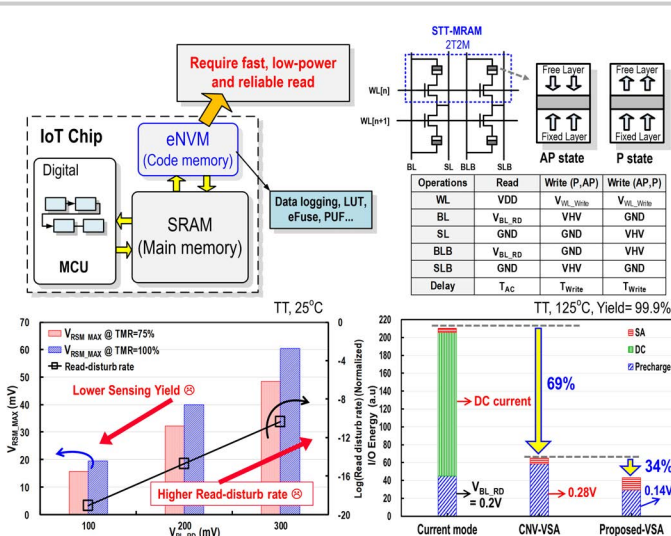


Figure 30.3.1: 2T2MTJ applications and the read challenge.

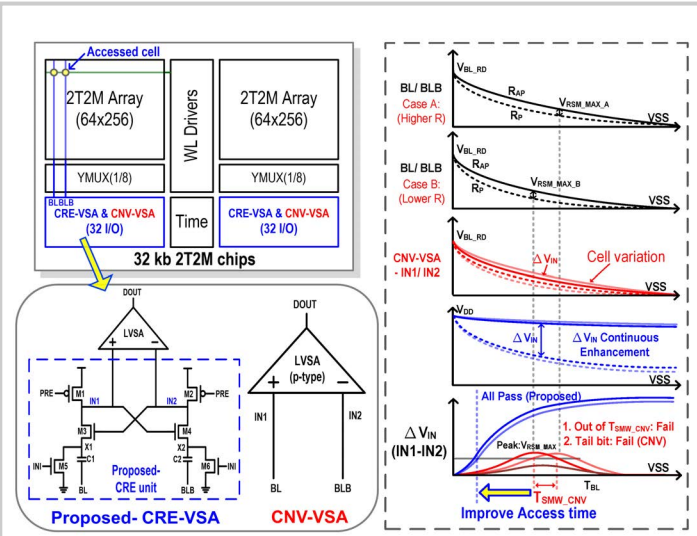


Figure 30.3.2: Concept and circuit of proposed CRE-VSA.

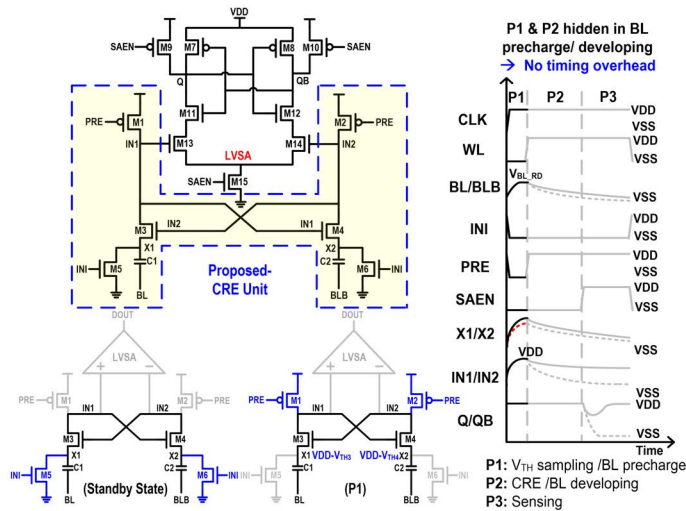


Figure 30.3.3: Continuous-recording-and-enhancement voltage sense amplifier operation - part 1.

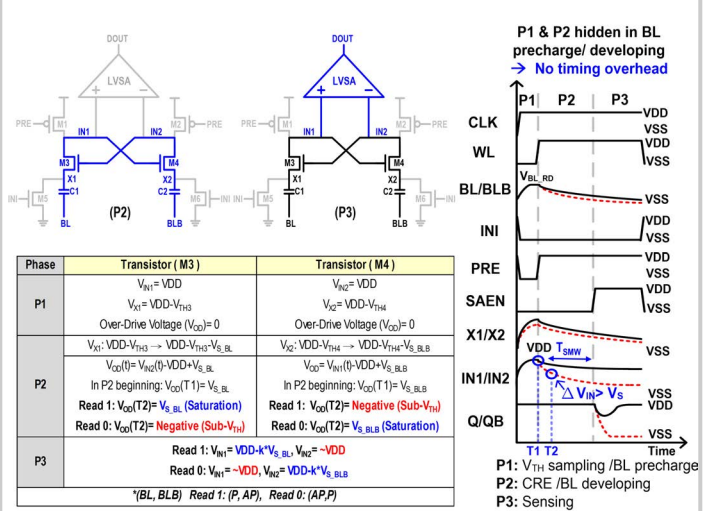


Figure 30.3.4: Continuous-recording-and-enhancement voltage sense amplifier operation - part 2.

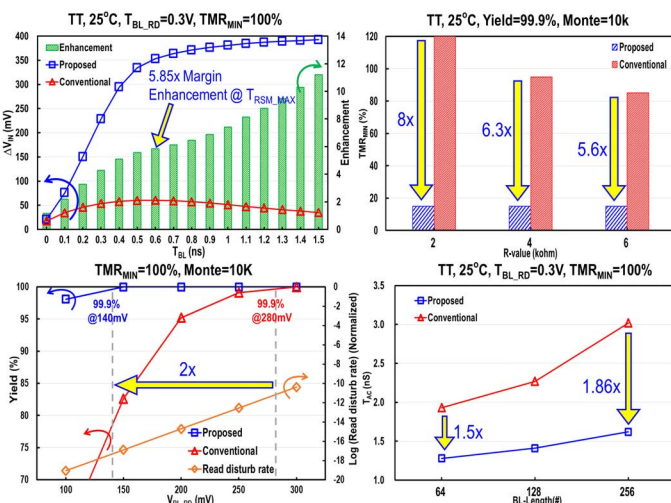


Figure 30.3.5: Continuous-recording-and-enhancement voltage sense amplifier performance summary.

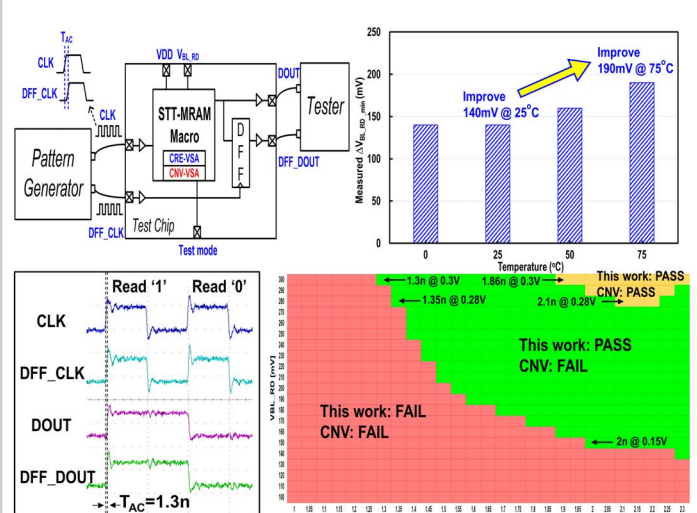


Figure 30.3.6: Measured results.

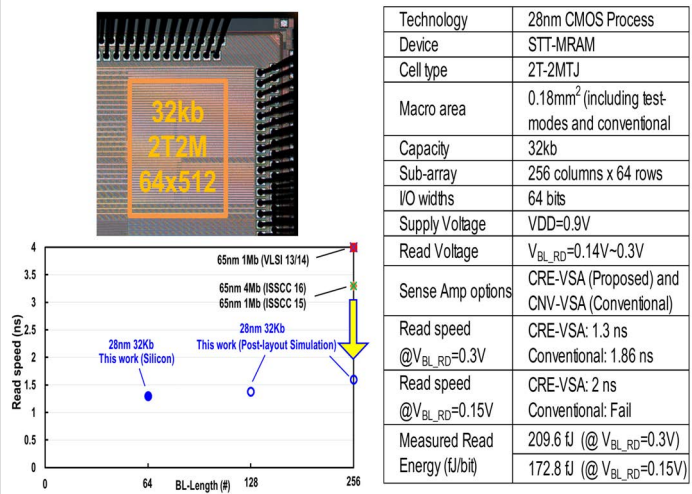


Figure 30.3.7: Die photo and summary table.